COLORADO STATE UNIVERSITY

Forecasting the Future: *Corn-for-Grain Yields*

By Team 12 : Will Code for Corn 🌽❤️

DevNet Hackathon 2026

Meet Our Team

Maxwell Freeman

Computer Engineer, Freshmen

I wanted to experience some high level programming and challenge myself.

Catalina Alcaraz

Computer Science, Soph.

I wanted to try something new and challenge myself by working on real world problems, plus I got to meet new people!

Jacob Auman

Mech Engr, Graduating

I joined the hackathon because I love working on challenging problems!

Jacob Christensen

Computer Sci, Soph/Jr

Meet a few people, learn a few things

SUSTAINABLE DEVELOPMENT GOALS

2 ZERO HUNGER

THE SUSTAINABLE DEVELOPMENT GOALS REPORT 2023: SPECIAL EDITION- UNSTATS.UN.ORG/SDGS/REPORT/2023/

Source: https://sdgs.un.org/goals/goal2

Problem Statement

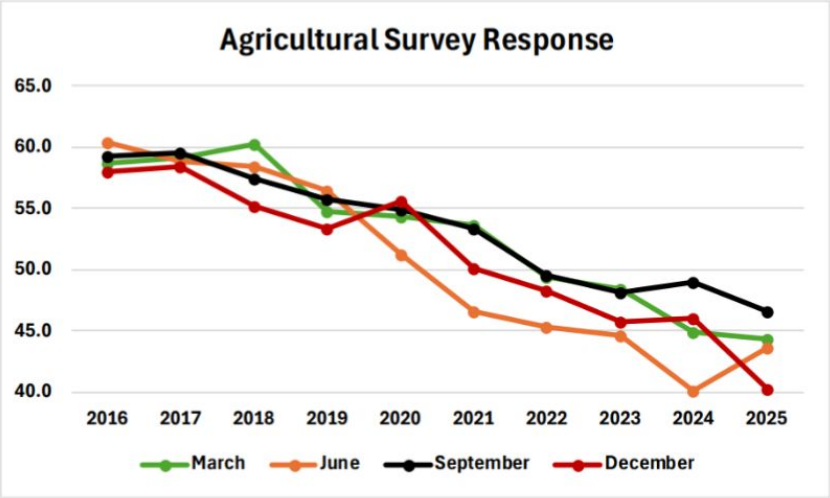
Create an efficient ML model that can accurately predict corn yields without the need of surveys every growing season.

We are aiming to develop this model such that it can accurately predict the amount of corn that will be yielded in a give year, allowing the NASS to save significant funds and time.

Stakeholder Interview Results

Currently the USDA NASS needs:

- ~1,600 personnel needed in field.
- This cost $1,500,000 $ per survey,
- **Four times a year** recurring cost and scale.



But this is an operational risk, as participation declines year over year

The statistical power of each estimate will decrease.

“Every 5 years, for the Census of Agriculture we send out surveys to nearly 3 Million Farmers

…

~ 40% don’t respond.”

Our Solution

Benefits

- Using our model will allow the USDA and NASS to save significant funds and time
- Companies can use this to adjust their pricing for the amount of corn thats going to be available
- Farmers can use the predicted price to estimate profit margins and budget accordingly

Features

- Estimates yearly corn yields using local weather and climate data, as well as Saturn satellite imagery
- Training based on state to improve predictions across large geographic areas.
- Extensible model allows for easy integration of additional datasets

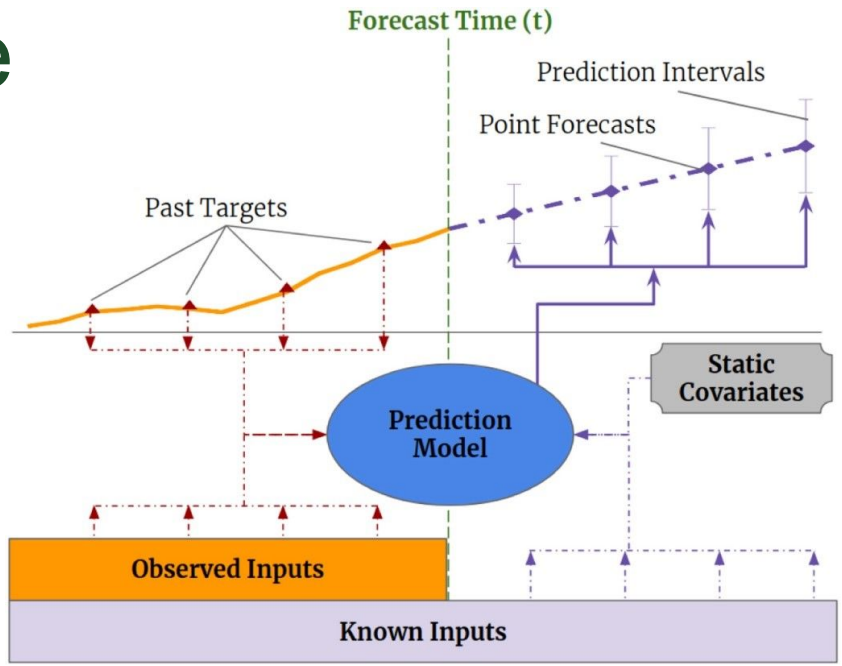
Model Architecture

When looking at the shape of the challenge and the data available we identified key features:

- Long Historical Record at fixed intervals
- Spatial Groupings of each county within each state.

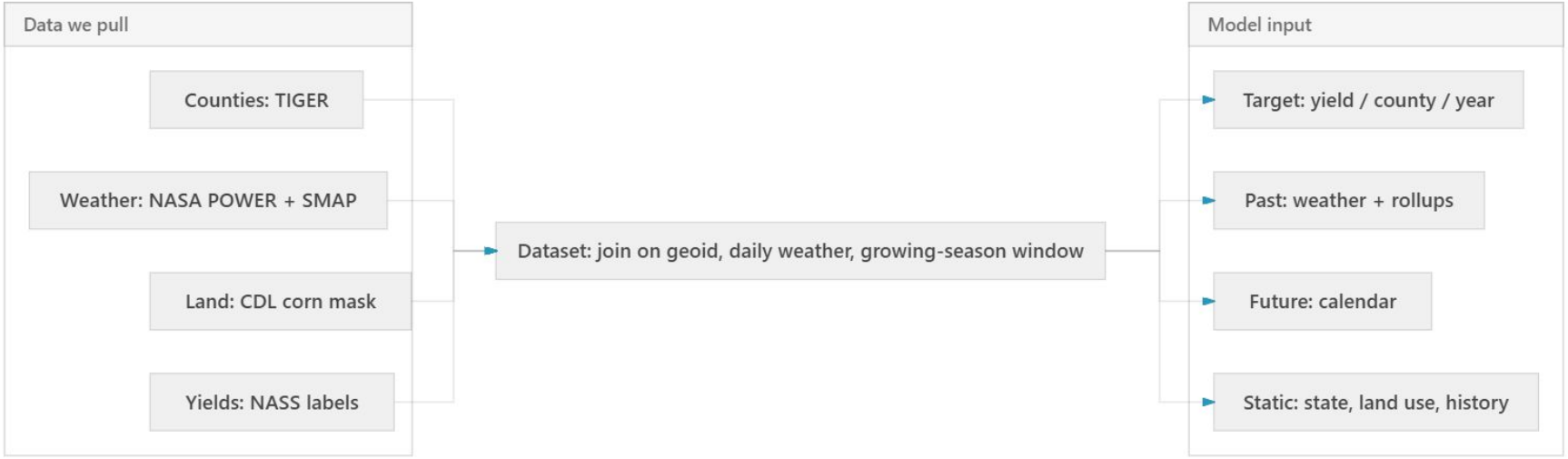
Temporal Fusion Transformer

Was designed by Google Researchers to address this exact style of problem.



Lim, Bryan, et al. "Temporal fusion transformers for interpretable multi-horizon time series forecasting." *International journal of forecasting* 37.4 (2021): 1748-1764.

Pipeline Architecture



Results

	2025 Year		August 2025		September 2025		October 2025 (no NASS data)		November 2025	
	Truth	Prediction	Truth	Prediction	Truth	Prediction			Truth	Prediction
Colorado	133		118		126				130	
Nebraska	194		192		191				191	
Wisonsin	188		185		184				183	
Missouri	185		191		184				184	
Iowa	210		222		219				216	

Results

state_fips	state_name	forecast_date	n_counties	state_yield_p10	state_yield_p50	state_yield_p90
8	Colorado	1-Aug	64	*110.33*	*126.94*	*146.69*
8	Colorado	final	64	*119.60*	*135.50*	*151.12*
8	Colorado	1-Oct	64	*119.61*	*139.06*	*154.90*
8	Colorado	1-Sep	64	*118.43*	*137.22*	*154.80*
19	Iowa	1-Aug	99	*175.69*	*187.80*	*199.18*
19	Iowa	final	99	*165.48*	*176.32*	*187.70*
19	Iowa	1-Oct	99	*190.58*	*201.12*	*212.36*
19	Iowa	1-Sep	99	*186.51*	*198.19*	*209.38*
29	Missouri	1-Aug	115	*147.15*	*160.08*	*171.53*
29	Missouri	final	115	*140.09*	*152.75*	*166.83*
29	Missouri	1-Oct	115	*164.97*	*177.41*	*189.74*
29	Missouri	1-Sep	115	*163.44*	*176.88*	*189.10*
31	Nebraska	1-Aug	93	*164.17*	*176.54*	*188.70*
31	Nebraska	final	93	*157.11*	*170.56*	*182.55*
31	Nebraska	1-Oct	93	*173.98*	*186.82*	*199.49*
31	Nebraska	1-Sep	93	*174.31*	*187.64*	*200.75*
55	Wisconsin	1-Aug	72	*164.26*	*175.07*	*185.53*
55	Wisconsin	final	72	*156.49*	*167.32*	*178.52*
55	Wisconsin	1-Oct	72	*171.31*	*181.96*	*191.66*
55	Wisconsin	1-Sep	72	*168.80*	*180.28*	*190.73*

Harmonized Landsat Sentinel-2 Findings

Timing the Search:

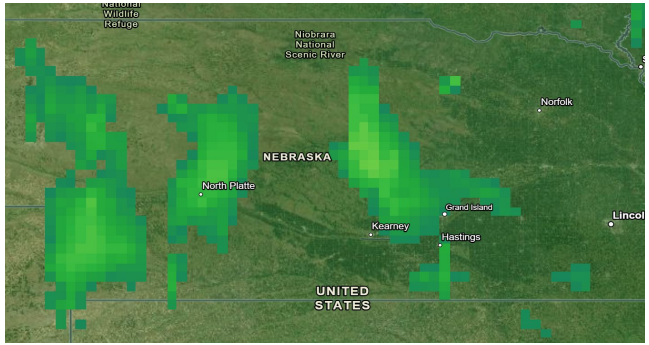
Prioritizing biological growing seasons to target the exact **2-week** windows—like July pollination—when corn is most sensitive to stress. This helps ignore seasonal noise and "pay attention" to the moments that actually determine harvest volume.

Translating Pixels to Data:

Since raw satellite photos are too complex for a model to read directly, by using Prithvi to translate images into "digital fingerprints." This can distill millions of pixels into 64 key features, giving the TFT a clear mathematical summary of crop health.

Connecting the Dots:

Satellites are often blocked by clouds, leaving "holes" in the data. Ideally we could fill these gaps by connecting the dots between snapshots, turning scattered images into a smooth daily timeline so the TFT can track the crop’s progress without interruption.



NASA EARTH DATA

Personal Takeaways

- Great tools exist, don’t be intimidated by hard problems
- Don't be afraid to learn new tools, keep expanding your capabilities to solve those hard problems
- Asking questions is critical, you can't get anywhere if you dont know your stakeholders or the problem
- Use every resource you can find, anything can be useful
- Its important to rest in order to maintain a productive work environment
- Even if you don’t get the exact expected result, you have to keep trying